

Executive Business Report

End-to-End Sales Forecasting & Demand Intelligence System

Executive Summary

This project presents an end-to-end Sales Forecasting and Demand Intelligence System developed using Python, Machine Learning, and Time Series Analysis techniques. The objective was to analyze historical retail sales data, predict future product demand, identify unusual sales patterns, segment products based on demand behavior, and present insights through an interactive Streamlit dashboard. The system helps businesses improve inventory planning, reduce stock shortages and overstock situations, and support data-driven decision-making.

Key Findings from Exploratory Data Analysis (EDA)

- The sales data covers four years of retail transactions across multiple product categories and regions.
 - Technology products generated the highest overall revenue, followed by Office Supplies and Furniture.
 - Sales showed a steady upward trend over the years, indicating business growth.
 - Monthly sales analysis revealed seasonal patterns with higher sales during festive and holiday periods.
 - Average shipping time remained consistent across most regions, indicating stable logistics performance.
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Sales Forecast

Three forecasting models were developed and compared:

- **SARIMA** – Statistical time series forecasting model.
- **Facebook Prophet** – Business forecasting model with automatic trend and seasonality detection.
- **XGBoost** – Machine learning model using lag features and rolling statistics.

Based on evaluation metrics (MAE, RMSE, and MAPE), **XGBoost achieved the best forecasting performance** and was selected as the production model.

The model predicts sales for the next three months, enabling proactive inventory management and better demand planning.

Anomaly Detection Results

Two anomaly detection techniques were implemented:

- Isolation Forest
- Z-Score Analysis

The system successfully detected unusual sales spikes and sudden drops.

Possible reasons include:

- Festival and holiday sales
- Promotional campaigns
- Supply chain disruptions
- Seasonal demand fluctuations

These anomalies help businesses investigate unexpected events and improve operational planning.

Product Demand Segmentation

K-Means Clustering was applied to segment products based on:

- Total Sales
- Growth Rate
- Sales Volatility
- Average Order Value

Products were grouped into four demand categories:

- High Volume, Stable Demand
- Growing Demand
- Low Volume, High Volatility
- Declining Demand

These segments help optimize stocking strategies and inventory allocation.

Business Recommendations

1. **Increase inventory** for products in the High Volume and Growing Demand segments to avoid stock shortages.
2. **Monitor anomalies regularly** to identify unusual sales patterns caused by promotions, seasonal events, or operational issues.

3. **Use XGBoost forecasts** for monthly demand planning, purchasing decisions, and warehouse management to improve forecasting accuracy.
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Limitations

- The forecasting model is trained only on historical sales data and does not consider external factors such as weather, inflation, competitor pricing, or marketing campaigns.
 - Forecast accuracy depends on data quality and historical patterns.
 - Predictions may become less accurate if customer behavior changes significantly over time.
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Conclusion

The Sales Forecasting and Demand Intelligence System successfully combines time series analysis, machine learning, anomaly detection, clustering, and interactive visualization into a single business solution. The system provides accurate sales forecasts, identifies unusual sales events, segments products by demand, and supports informed inventory and supply chain decisions. With additional external business data, the system can be further enhanced to deliver even more accurate and reliable demand forecasting for real-world retail applications.